

EDUCATION

UNSW Sydney | Bachelor of Commerce (Finance) | Aug 2023 - Expected Aug 2026

Core coursework: Trading & Market Making; Data & Algorithms in Trading; Derivatives & Risk; Financial Market Data Design and Analysis; Numerical Methods; Mathematical Economics; Data & ML.

Project results: Python Financial Data & Portfolio Analysis - **96.25/100** individual; **92.71/100** team.

Scholarship: UNSW International Student Award - **20%** tuition award.

SIG ALGOTHON 2026

1st

LIVE LEADERBOARD

SCORE 984.62 | 3 AUG 2026

TECHNICAL SKILLS

PROGRAMMING & DATA

Python, SQL/SQLite, FastAPI; pandas, NumPy, scikit-learn, Jupyter; basic C++.

QUANTITATIVE TRADING

L2 order books, microprice, IOC/passive quoting, market making, execution, backtesting, inventory/risk controls.

AI & AGENTS

RAG, LangGraph, PEFT/QLoRA, MCP, LLM evaluation, multi-agent simulation, AI safety.

SYSTEMATIC QUANTITATIVE TRADING

Market Microstructure & Intraday Cross-Asset Trading

Feb 2025 - Jul 2026

- Team Lead & Quantitative Researcher
- Reconstructed trade-aligned L2 snapshots from nanosecond ASX/Cboe prints and quotes using same-venue, strictly prior timestamps; filtered auctions, stale/crossed quotes, zero-depth levels and abnormal spreads, retaining **5,953 validated matches**.
 - Built 5-second ETF cross-asset fair-value signals and 1-minute/1-second equity microstructure features using Nasdaq 100 futures, AUD/USD, spread, depth, order imbalance, volatility and time-of-day effects to identify residual dislocations and adverse selection.
 - Ran chronologically split, cost-aware simulations with next-bar fill proxies, fees, slippage, inventory limits and end-of-day liquidation; used ablation, stress testing and P&L attribution to separate signal value from execution assumptions.

Algorithmic Trading, Market Making & Execution

May 2026 - Present

- Algorithmic Trading Developer
- Connected to the Optibook Exchange API for five-level books, positions and P&L; built an automated ASML/ASML_DUAL loop using equal-volume IOC legs for crossed-book spreads and inventory-aware passive quotes.
 - Implemented quote refresh, one-leg/self-trade protection, +/-100-lot limits, EWMA-volatility adaptive thresholds, covariance risk gating, fill confirmation, reconnection and automatic flattening.
 - A saved one-hour simulation generated **+34.80 P&L, 1,142 lots and 378 fills**, ending flat; a next-trade model trained on 3,299 observations reduced RMSE by **4.37%** versus the DUAL mid-price benchmark.

SIG Algothon Systematic Multi-Asset Strategy

Jun 2026 - Present

- Team Captain & Quant Strategy Developer
- Extended a lead-lag/reversal core into a four-horizon pair portfolio calibrated on full-history, 500-, 420- and 360-day windows; combined ADF, AR(1), correlation and mean-crossing filters with 100/120/150-day z-score consensus, hysteresis and voting.
 - Designed dollar caps, beta-drift, turnover and active-pair constraints for 51 instruments, limiting the book to 10 pairs; preserved the highest-conviction signals and tested extensions under 10 bps commissions and tail-loss controls.
 - Locked code and configuration hashes, reconciled local traces with the official evaluator and ran rolling 250/100/50-day stability checks; the team ranked **1st** on the [official live Algothon 2026 leaderboard](#) with a score of **984.62** as of 3 Aug 2026.

AI QUANTITATIVE RESEARCH

Point-in-Time Multi-Asset, News & Event Pricing

Jul 2026 - Present

- AI Quantitative Researcher
- Built point-in-time panels for **50 US equities, 10 cryptocurrencies and 149,683 news headlines**, with ETL, deduplication, source provenance, timezone, exchange-calendar and publication-time alignment.
 - Produced reproducible daily market/news panels and return features for out-of-sample multi-asset and sentiment research, with publication-time controls to prevent leakage.
 - Configured Modal/FastAPI, signed webhooks and idempotent event handling for Explaining Markets; developing an earnings-event agent that freezes point-in-time inputs and submits abnormal-return percentiles for live out-of-sample evaluation.

Evidence-Grounded Financial Agents & AI Engineering

Jul 2026

- AI Mockathon | UNSW FAIC AI Engineering Training
- Built a bounded financial agent restricted to RBA, ASX and AFR sources, using deterministic queries, structured tool traces and evidence-constrained synthesis that links sources, parameters and intermediate outputs to conclusions.
 - Developed an assistive pricing workflow combining market signals, Gemini-generated explanations and deterministic rules; evaluated disagreement, uncertainty, drift, user autonomy and fairness rather than treating agent outputs as ground truth.
 - Completed UNSW-delivered AI engineering training under FAIC; applied open/self-hosted models, RAG, LangGraph, MCP, multi-agent workflows and 4-bit QLoRA adaptation of Llama-3.1-Nemotron-Nano-8B (r=32, alpha=64).

MULTI-AGENT SIMULATION & AI SAFETY

Multi-Agent Behaviour, Game Theory & AI Safety

Nov 2025 - Present

- Research Assistant, UNSW Sydney
- Built a research-grade public agent-network dataset spanning **1,548,232 posts, 13,389,884 comments, 762,644 profiles and 32,081 communities**; implemented resumable collection, rate-limit handling, incremental deduplication and integrity checks, identifying 86,194 human-claimed agents.
 - Developing game-theoretic multi-agent experiments that vary information structure, memory, repeated interaction and auction/market rules to distinguish autonomous strategy from human prompting, scripted automation and identity ambiguity.
 - Designing counterfactual interventions for reward misspecification, unsafe exploration, distribution shift, collusion and correlated failure, using trace attribution, ablation and simulation-validity checks to test robustness and empirical-to-simulation transfer.

PittsAI - Multi-Agent Customer Simulation

Nov 2025 - Jul 2026

- Founder & Research Lead
- Built a pre-seed customer-agent simulation prototype that converts company materials and user-defined operating assumptions into comparable scenarios for product, pricing, channel and service decisions.
 - Varied customer types, behavioural rules and business assumptions in controlled scenarios, logged output differences and retained human review so synthetic results were not treated as direct market evidence.
 - Selected as a **Peter Farrell Cup 2026 finalist** and for **Antler Australia AUS16**; used customer interviews, co-founder matching and mentor feedback to narrow product scope, positioning and go-to-market assumptions.